Solving the Simplified Model, Step by Step

This document derives the solution of the simplified model, one substitution at a time. It adds no assumptions and no results to the model section. Its purpose is to show how the equilibrium conditions reduce to a small system that can be solved, which objects have closed forms, and which ones must be computed. Propositions and theorems cited by name and number, and all notation, are those of the model section, whose notation table lists every symbol.

The model has quantity competition among groups, Cobb–Douglas preferences across sectors, labor as the only input, non-negative tariffs, and plants that serve a given set of markets. Section 8 adds the extension in which groups choose which plants serve each market.

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0 What Has to Be Solved

Primitives. The economy is described by the sets of countries $\mathcal{N}$ and sectors $\mathcal{K}$; labor endowments L_n; spending shares β_k and elasticities $\sigma_k > 1$; the groups, with their headquarters, sectors and plants; plant productivities φ_i; the markets each plant serves; trade costs $d_{\ell n}$; tariffs $t_{\ell nk} \geq 0$; and ownership shares θ_{gn}. All of these are given.

Unknowns. An equilibrium consists of wages w_n, incomes X_n, and the price $p_{i,n}$ and quantity $q_{i,n}$ of every plant in every market it serves. Market sizes E_{nk}, price indices P_{nk}, shares, markups and profits follow from these.

Strategy. There is no closed form for the equilibrium as a whole. Every object, however, can be written as a function of the wage vector w alone, through a chain of substitutions in which each link is either a closed form or a problem with a unique and easily computed solution. The equilibrium is then a wage vector that satisfies $N-1$ labor-market conditions.

Step	Solves for	Takes as given	How
1	demand functions	prices, spending	closed form
2	unit and delivered costs	wages	closed form
3	shares, markups, prices in one market	delivered costs	one scalar equation
4	market sizes and incomes	wages	linear system
5	labor-market conditions	everything above	closed form
6	wages	—	$N-1$ equations

Steps 1–3 hold at any wages. Market shares do not depend on spending, so Step 3 can be solved before Step 4, and incomes then solve a linear system.

1 Step 1: Household Demand

Within a sector. Fix country n and sector k, and write $\sigma = \sigma_k$. To obtain an amount C_{nk} of the sector's bundle at least cost, the household chooses $q_{i,n}$ to minimize $\sum_{i \in \mathcal{A}_{nk}} p_{i,n} q_{i,n}$ subject to $\big[\sum_i q_{i,n}^{(\sigma-1)/\sigma}\big]^{\sigma/(\sigma-1)} = C_{nk}$. With multiplier ξ, the first-order condition for variety i is

$$p_{i,n} = \xi \, \frac{\partial C_{nk}}{\partial q_{i,n}} = \xi \, C_{nk}^{1/\sigma} q_{i,n}^{-1/\sigma}, \qquad \text{so} \qquad q_{i,n} = \Big(\frac{p_{i,n}}{\xi}\Big)^{-\sigma} C_{nk}.$$

Substituting into the constraint, $C_{nk} = \xi^{\sigma} C_{nk} \big[\sum_i p_{i,n}^{1-\sigma}\big]^{\sigma/(\sigma-1)}$, so $\xi = \big[\sum_i p_{i,n}^{1-\sigma}\big]^{1/(1-\sigma)} \equiv P_{nk}$. The multiplier is the price index: the cost of one unit of the bundle. Spending on the sector is then $P_{nk} C_{nk}$, and

$$q_{i,n} = p_{i,n}^{-\sigma} P_{nk}^{\sigma-1} \, (P_{nk} C_{nk}). \tag{1}$$

Across sectors. The household maximizes $U_n = \prod_k C_{nk}^{\beta_k}$ subject to $\sum_k P_{nk} C_{nk} = X_n$. With multiplier ζ, the first-order condition for sector k is $\beta_k U_n / C_{nk} = \zeta P_{nk}$, so $P_{nk} C_{nk} = \beta_k U_n / \zeta$. Adding over sectors and using $\sum_k \beta_k = 1$ gives $U_n / \zeta = X_n$, hence

$$E_{nk} \equiv P_{nk} C_{nk} = \beta_k X_n. \tag{2}$$

Each sector receives a fixed share of income. Substituting into (1),

$$q_{i,n} = p_{i,n}^{-\sigma} P_{nk}^{\sigma-1} E_{nk}, \qquad s_{i,n} \equiv \frac{p_{i,n} q_{i,n}}{E_{nk}} = \left(\frac{p_{i,n}}{P_{nk}}\right)^{1-\sigma}. \tag{3}$$

The revenue shares $s_{i,n}$ add up to one in every market. Substituting the optimal bundles into utility gives $U_n = X_n/P_n$ with $P_n = \prod_k (P_{nk}/\beta_k)^{\beta_k}$, which is why welfare is real income.

2 Step 2: Costs

Unit cost. A plant located in ℓ produces φ_i units of output per unit of labor. Its cost of producing one unit is

$$a_i = \frac{w_\ell}{\varphi_i}. \tag{4}$$

Delivered cost. To sell $q_{i,n}$ units in n, the plant produces $d_{\ell n} q_{i,n}$ units, because of iceberg trade costs. Its cost of serving n, exclusive of tariffs, is

$$V_{i,n} = a_i d_{\ell n} q_{i,n}, \tag{5}$$

country n collects $t_{\ell nk} V_{i,n}$ in tariffs, and the cost of delivering one more unit to n, tariff included, is

$$c_{i,n} = a_i \, d_{\ell n} \, (1 + t_{\ell nk}) = \frac{w_\ell \, d_{\ell n} (1 + t_{\ell nk})}{\varphi_i}. \tag{6}$$

Where the money goes. The whole of $V_{i,n}$ is paid to labor in ℓ. Out of the plant's revenue in n, what is left after $V_{i,n}$ and the tariff is profit (Step 3).

3 Step 3: One Market

Fix a market (n, k) and drop its indices: σ, E, P, c_i, s_i and $t_i \equiv t_{\ell(i)nk}$. Let T_g be the plants of group g in the market. The inputs of this step are the delivered costs c_i, which by (6) depend only on wages, and the spending level E. The outputs are prices, quantities, shares, markups and profits.

3.1 Inverse demand

Let $\gamma = (\sigma - 1)/\sigma$ and $Q = (\sum_i q_i^\gamma)^{1/\gamma}$, the quantity of the sector's bundle. Since $PQ = E$, (1) can be inverted as $p_i = P(q_i/Q)^{-1/\sigma} = (E/Q)(q_i/Q)^{-1/\sigma}$, so that

$$p_i = E \, Q^{-\gamma} q_i^{\gamma-1}, \qquad s_i = \frac{p_i q_i}{E} = \frac{q_i^\gamma}{Q^\gamma}. \tag{7}$$

3.2 The group's first-order condition and the markup

Group g chooses q_i for every $i \in T_g$ to maximize $\pi_g = \sum_{i \in T_g} (p_i - c_i) q_i$. Its revenue is

$$\sum_{j \in T_g} p_j q_j = E \, Q^{-\gamma} \sum_{j \in T_g} q_j^\gamma = E \, S_g, \qquad S_g \equiv \sum_{j \in T_g} s_j.$$

Using $\partial \ln Q/\partial q_i = q_i^{\gamma-1}/Q^\gamma = s_i/q_i$, the marginal revenue of plant i is

$$\frac{\partial(ES_g)}{\partial q_i} = E\,Q^{-\gamma}\Big[\gamma\, q_i^{\gamma-1} - \gamma\,\frac{q_i^{\gamma-1}}{Q^\gamma}\sum_{j\in T_g} q_j^\gamma\Big] = \gamma\, p_i\,(1-S_g).$$

The first term is the revenue from selling more. The second is the sales that the expansion takes from all of the group's plants, plant i included, because selling more lowers the price index. A single-plant firm counts only its own loss, which gives the factor $1-s_i$; a group also counts the loss on its other plants, which is cannibalization, so the factor is $1-S_g$. Setting marginal revenue equal to marginal cost c_i gives

$$p_i = \mu(S_g)\,c_i, \qquad \mu(S) = \frac{\sigma}{(\sigma-1)(1-S)}. \tag{8}$$

The markup is the same for every plant of the group and depends only on the group's total share. It starts at $\mu(0) = \sigma/(\sigma-1)$, the markup of a negligible firm, and is unbounded as $S \to 1$, which is why every market needs at least two groups.

3.3 Shares within a group

With a common markup, $p_i/p_j = c_i/c_j$ for any two plants of the same group, so by (3) $s_i/s_j = (c_i/c_j)^{1-\sigma}$. Adding over the group,

$$\frac{s_i}{S_g} = \frac{c_i^{1-\sigma}}{K_g}, \qquad K_g \equiv \sum_{i\in T_g} c_i^{1-\sigma}. \tag{9}$$

The group gives its cheaper plants a larger part of its sales. Costs show up in how the group splits its sales, not in its markups.

3.4 The share equation

Let $A \equiv \sum_i p_i^{1-\sigma} = P^{1-\sigma}$, so that $s_i = p_i^{1-\sigma}/A$. Substituting $p_i = \mu(S_g)c_i$ and adding over the group,

$$S_g = \sum_{i\in T_g} \frac{\mu(S_g)^{1-\sigma} c_i^{1-\sigma}}{A} = \frac{\mu(S_g)^{1-\sigma} K_g}{A},$$

that is,

$$\psi(S_g) = \frac{K_g}{A}, \qquad \psi(S) \equiv S\,\mu(S)^{\sigma-1} = \Big(\frac{\sigma}{\sigma-1}\Big)^{\sigma-1}\frac{S}{(1-S)^{\sigma-1}}. \tag{10}$$

For a given A this is one equation in one unknown for each group. Taking logs,

$$\ln S_g - (\sigma-1)\ln(1-S_g) = \ln K_g - \ln A - (\sigma-1)\ln\frac{\sigma}{\sigma-1} \equiv x_g(A). \tag{11}$$

The left side rises from $-\infty$ to $+\infty$ as S_g goes from 0 to 1, so the equation has exactly one root $S_g(A) \in (0,1)$ for every $A > 0$. It has no closed form in general. The left side is not concave, so a plain Newton step can leave $(0,1)$; Newton's method with a bisection step whenever a Newton step would leave the current bracket converges quickly, using the derivative of the left side, $[1+(\sigma-2)S]/[S(1-S)]$.

3.5 The market-clearing equation

Shares must add up to one, so the market's equilibrium is the value of A that solves

$$\mathcal{S}(A) \equiv \sum_g S_g(A) = 1. \tag{12}$$

Differentiating (10) in logs, with $d\ln\psi/d\ln S = 1 + (\sigma-1)S/(1-S)$,

$$\frac{dS_g}{d\ln A} = -S_g\Lambda_g, \qquad \Lambda_g \equiv \frac{1-S_g}{1+(\sigma-2)S_g} > 0. \tag{13}$$

Every share falls as A rises, so $\mathcal{S}$ is strictly decreasing and (12) has at most one solution (Proposition 2). A bracket that contains it is found as follows. Since $\mu(S) \geq \mu(0)$, (10) gives $S_g \leq K_g\mu(0)^{1-\sigma}/A$, so at $A_{\text{hi}} = \mu(0)^{1-\sigma}\sum_g K_g$ shares add up to at most one. As $A \to 0$ every share tends to one, so $\mathcal{S}$ tends to the number of groups, which is at least two; halving A from A_{hi} reaches a point A_{lo} with $\mathcal{S}(A_{\text{lo}}) \geq 1$ in finitely many steps. Between the two, Newton's method on $\ln A$, with derivative $\mathcal{S}'(\ln A) = -\sum_g S_g\Lambda_g$ and a bisection step whenever a Newton step would leave the bracket, converges to the unique root.

3.6 Recovering everything else

Once A and the shares S_g are known, every other object in the market follows in closed form:

$$\mu_g = \mu(S_g), \qquad p_i = \mu_g c_i, \qquad P = A^{1/(1-\sigma)}, \tag{14}$$

$$s_i = S_g\,\frac{c_i^{1-\sigma}}{K_g}, \qquad r_i = s_i E, \qquad q_i = \frac{r_i}{p_i}, \tag{15}$$

$$V_i = \frac{r_i}{\mu_g(1+t_i)}, \qquad \text{tariff revenue } t_iV_i. \tag{16}$$

The price index is consistent with the prices, since $\sum_i p_i^{1-\sigma} = \sum_g \mu_g^{1-\sigma}K_g = A\sum_g S_g = A$. Profit is

$$\pi_g = \sum_{i\in T_g}(p_i - c_i)q_i = \Big(1 - \frac{1}{\mu_g}\Big)\sum_{i\in T_g} r_i = E\,S_g\,m(S_g), \qquad m(S) \equiv 1 - \frac{1}{\mu(S)} = \frac{1+(\sigma-1)S}{\sigma},$$

so that, with $\kappa = \sigma - 1$,

$$\pi_g = \frac{E}{\sigma}(S_g + \kappa S_g^2). \tag{17}$$

The revenue split. For each plant, revenue splits exactly into labor cost, tariff and profit:

$$r_i = \mu_g(1+t_i)V_i = \underbrace{V_i}_{\text{labor}} + \underbrace{t_iV_i}_{\text{tariff}} + \underbrace{(\mu_g - 1)(1+t_i)V_i}_{\text{profit}}. \tag{18}$$

Dividing by E and adding over plants, each dollar spent in the market splits as

$$\underbrace{\sum_i \frac{s_i}{\mu_{g(i)}(1+t_i)}}_{v\text{: labor}} + \underbrace{\sum_i \frac{t_i s_i}{\mu_{g(i)}(1+t_i)}}_{\tau\text{: tariffs}} + \underbrace{\sum_g S_g\,m(S_g)}_{\text{profit}} = 1. \tag{19}$$

Two properties used below. First, shares, markups and the price index depend only on delivered costs, not on E: spending scales quantities, revenue, costs and profits, and nothing else. Second, multiplying every delivered cost by the same factor leaves every share and markup unchanged and multiplies every price and P by that factor. Since delivered costs are proportional to wages, scaling all wages scales all prices and leaves all shares and markups unchanged.

4 Step 4: Market Sizes and Incomes

From now on wages w are given. By Steps 2 and 3, every delivered cost, share and markup is then known, and everything that remains is linear in spending.

Profits and tariffs as a function of spending. Group g of sector k earns $E_{n'k}S_{g,n'}m(S_{g,n'})$ in each destination n' it serves, and country n collects $TR_n = \sum_k \tau_{nk}E_{nk}$ in tariffs, where τ_{nk} is the tariff part of (19) in market (n,k). With $\sum_n \theta_{gn} = 1$, the income of country n is

$$X_n = w_n L_n + \sum_g \theta_{gn} \sum_{n'} E_{n'k(g)} S_{g,n'} m(S_{g,n'}) + \sum_k \tau_{nk} E_{nk}.$$

Substituting market sizes. By (2), $E_{n'k} = \beta_k X_{n'}$. Substituting,

$$X_n = w_n L_n + \sum_{n' \in \mathcal{N}} \Gamma_{nn'} X_{n'}, \qquad \Gamma_{nn'} \equiv \sum_k \beta_k \Big[\sum_{g:\, k(g)=k} \theta_{gn} S_{g,n'} m(S_{g,n'}) + \mathbf{1}\{n' = n\}\, \tau_{nk} \Big]. \quad (20)$$

$\Gamma_{nn'}$ is the income that country n receives, as profit or tariff revenue, out of each dollar that country n' spends. In matrix form,

$$X = wL + \Gamma X \qquad \Longrightarrow \qquad X = (I - \Gamma)^{-1} wL. \quad (21)$$

Why the solution exists, is unique and exceeds labor income. Every entry of Γ is non-negative, because profits and tariffs are. The column of Γ for country n' adds up, over the receiving countries n, to

$$\sum_n \Gamma_{nn'} = \sum_k \beta_k \Big[\sum_g S_{g,n'} m(S_{g,n'}) + \tau_{n'k} \Big] = 1 - \sum_k \beta_k v_{n'k} < 1,$$

by (19): of each dollar country n' spends, the part that goes to labor never comes back as profit or tariff revenue. A non-negative matrix whose columns add up to less than one has all its eigenvalues below one in absolute value, so

$$(I - \Gamma)^{-1} = I + \Gamma + \Gamma^2 + \cdots \geq 0, \qquad X = wL + \Gamma wL + \Gamma^2 wL + \cdots \geq wL.$$

The series has a simple reading. Labor income wL is spent; part of that spending returns as profits and tariffs, ΓwL; that income is spent again, and so on. Every country's income is at least its labor income (Proposition 3). Market sizes follow from (2), and with them every quantity, cost and profit through (15)–(17).

Scale. Scaling all wages by a common factor leaves Γ unchanged and scales wL, so it scales incomes and market sizes by the same factor.

5 Step 5: The Labor Market

The wage bill. Country n's wage bill is the labor cost of all plants located in n, in every market they serve. Let

$$v_{n,(n',k)} \equiv \sum_{i \in \mathcal{A}_{n'k}:\, \ell(i)=n} \frac{s_{i,n'}}{\mu_{g(i),n'}(1+t_{nn'k})} \tag{22}$$

be the share of spending in market (n',k) that is paid to labor in n. Then, using (2),

$$W_n(w) = \sum_{n' \in \mathcal{N}} \sum_{k \in \mathcal{K}} v_{n,(n',k)}\, \beta_k\, X_{n'}. \tag{23}$$

Every term is known from Steps 3 and 4. The excess demand for labor is

$$z_n(w) = \frac{W_n(w)}{w_n} - L_n, \tag{24}$$

and labor markets clear when $z_n = 0$ for every n.

Walras' law. One of these conditions is redundant, because at any wages

$$\sum_n w_n z_n(w) = \sum_n W_n - \sum_n w_n L_n = 0.$$

To see it, add the revenue split (18) over all plants and markets: world spending equals the world wage bill plus world tariff revenue plus world profits, $\sum_n X_n = \sum_n W_n + \sum_n TR_n + \sum_g \pi_g$. Adding the income equations over countries, with $\sum_n \theta_{gn} = 1$, gives $\sum_n X_n = \sum_n w_n L_n + \sum_g \pi_g + \sum_n TR_n$. The two expressions are equal, so $\sum_n W_n = \sum_n w_n L_n$.

The numeraire. Scaling all wages by a common factor scales W_n and w_n by that factor (Steps 3 and 4), so z does not change. Only relative wages are determined, and we set $w_1 = 1$. By Walras' law, $z_1 = 0$ whenever $z_2 = \cdots = z_N = 0$.

6 Step 6: Wages

The reduced system. Steps 2–5 reduce the whole model to $N-1$ equations in $N-1$ unknowns:

$$z_n(1, w_2, \ldots, w_N) = 0, \qquad n = 2, \ldots, N. \tag{25}$$

Written out, the condition for country n is

$$w_n L_n = \sum_{n'} \sum_k \beta_k X_{n'} \sum_{i \in \mathcal{A}_{n'k}:\, \ell(i)=n} \frac{s_{i,n'}}{\mu_{g(i),n'}(1+t_{nn'k})}, \qquad X = (I-\Gamma)^{-1} wL,$$

where shares and markups depend on wages only through the delivered costs $c_{i,n'} = w_{\ell(i)} d_{\ell(i)n'}(1 + t_{\ell(i)n'k})/\varphi_i$. Evaluating $z(w)$ at a given w therefore means: compute delivered costs, solve every market (Step 3), solve the income system (Step 4), and compute the wage bills (Step 5).

Three channels. A rise in a foreign wage w_j affects country n's wage bill in three ways. Plants located in j become more expensive, so plants located in n gain sales, both within their own groups and against rival groups. The share changes move markups, which changes how much of each dollar of revenue is paid to labor. And incomes $X_{n'}$ change, because they include the profits of the groups each country owns. The last channel is the ownership channel of the model. Theorem 1 states when the first two always favor country n and the third does not undo them.

Damped tâtonnement. The algorithm updates wages by

$$w_n^{\text{new}} = w_n^{1-\lambda}\Big(\frac{W_n(w)}{L_n}\Big)^{\lambda}, \qquad \text{that is,} \qquad \ln w_n^{\text{new}} = \ln w_n + \lambda \ln\Big(1 + \frac{z_n(w)}{L_n}\Big), \tag{26}$$

with $\lambda \in (0,1)$, and then rescales so that $w_1 = 1$, which is allowed because the update is homogeneous of degree one. A country's wage rises when its labor is in excess demand, and a wage vector is a fixed point exactly when $W_n = w_n L_n$ for every n.

How much damping. Let $M_{nj} = \partial \ln W_n / \partial \ln w_j$. In logs, the update has derivative $(1-\lambda)I + \lambda M$. Its off-diagonal entries are positive when wages are gross substitutes, and its diagonal entries are positive when $\lambda < 1/(1 - M_{nn})$ for every n. When labor demand is elastic, $M_{nn} < 0$, the bound is below one, and an undamped update ($\lambda = 1$) can overshoot.

Newton steps. Close to the solution, Newton's method on $F_n(\ln w) = \ln W_n - \ln w_n - \ln L_n$, $n = 2, \ldots, N$, converges faster. Its derivative is $\partial F_n / \partial \ln w_j = M_{nj} - \mathbf{1}\{n = j\}$, for $n, j \geq 2$. The matrix M is obtained column by column, by changing one $\ln w_j$ slightly and re-evaluating Steps 2–5. Iterations stop when $\max_n |z_n|/L_n$ falls below a tolerance.

7 The Complete Algorithm, and What to Check

From the innermost step outward:

(A) *Costs.* Given w, compute a_i and $c_{i,n}$ by (4)–(6).

(B) *Markets.* In every market, solve each share equation (11) for $S_g(A)$ and the clearing equation (12) for A; recover μ_g, p_i, P and s_i by (14)–(15).

(C) *Incomes.* Build Γ and compute X by (21) and E by (2).

(D) *Labor market.* Compute W_n by (23) and z_n by (24).

(E) *Wages.* Update w by (26), then by Newton steps, repeating (A)–(D), until $z(w) = 0$.

Closed form	Computed
demand, unit and delivered costs	each group's share, given A (one monotone equation)
markups, prices, quantities given shares	each market's A (one monotone equation)
shares within a group, profits	incomes (one linear system)
revenue split into labor, tariffs, profit	wages ($N-1$ equations)
wage bills given incomes and shares	

Checks at the solution. Each of these holds at an equilibrium and is a useful test of any implementation.

1. In every market, shares add up to one and $\sum_i p_i^{1-\sigma} = P^{1-\sigma}$.
2. For every plant, revenue equals labor cost plus tariff plus profit, (18).
3. Every country's income equals labor income plus owned profits plus tariff revenue, and is at least its labor income.
4. Every labor market clears, and Walras' law holds also away from the solution, which tests the accounting at arbitrary wages.
5. The hypotheses of Theorem 1: no group holds more than half of any market, and $\bar{E}_n^{(j)} > 0$ for every pair of countries.

8 Extension: Endogenous Entry

With entry, each group chooses which of its potential plants $\mathcal{P}_g$ serve each market, and a plant that serves market (n,k) pays the fixed cost $F_{i,n} = f_k w_\ell f_{\ell n}^d$ in labor of its location. Steps 1–3 are unchanged for given plant sets. What changes is that plant sets are now unknowns.

8.1 Entry into one market

The value of a plant set. By Step 3, a set $T \subseteq \mathcal{P}_g$ affects the group's profit only through its strength $K(T) = \sum_{i\in T} c_i^{1-\sigma}$, and it costs $\Phi(T) = \sum_{i\in T} F_{i,n}$. Taking A as given, the group's share with set T is $\psi^{-1}(K(T)/A)$, and its net payoff is

$$U_g(T;A) = \Omega\big(K(T);A\big) - \Phi(T), \qquad \Omega(K;A) = \frac{E}{\sigma}(S+\kappa S^2)\Big|_{S=\psi^{-1}(K/A)}. \tag{27}$$

Serving the market with no plant gives $U_g = 0$.

The ladder. A group with $|\mathcal{P}_g|$ potential plants has $2^{|\mathcal{P}_g|}$ sets. Since Ω rises with K, a set can be chosen only if no other set has at least as much strength at no greater fixed cost. The construction is:

1. compute $(K(T), \Phi(T))$ for every set T;
2. discard every T for which some T' has $K(T') \geq K(T)$ and $\Phi(T') \leq \Phi(T)$, with one inequality strict;
3. order the remaining sets by strength; fixed cost rises along the order too. Call them rungs $0, 1, \ldots, R_g$, with rung 0 the empty set.

When all of a group's plants pay the same fixed cost F in the market, rung r is the set of the r plants with the lowest delivered costs. A stand-alone firm has two rungs: out or in.

The best response. At a given A, the group picks the rung with the highest payoff,

$$K_g^*(A) = K\big(T_g^*(A)\big), \qquad T_g^*(A) = \arg\max_{r=0,\ldots,R_g} \Omega(K_r;A) - \Phi_r. \tag{28}$$

With equal fixed costs this is a cutoff: the strength added by each rung falls as more expensive plants are added, and Ω is strictly concave in K (Theorem 2(a)), so the gain from one more rung falls, and the group climbs while the gain is positive.

The entry equilibrium of the market. Substituting the best responses into (12) gives one equation in A:

$$\mathcal{S}^*(A) \equiv \sum_g \psi^{-1}\Big(\frac{K_g^*(A)}{A}\Big) = 1. \tag{29}$$

When no group would hold more than half of the market with any of its rungs, each $K_g^*(A)$ does not rise with A (Theorem 2), so $\mathcal{S}^*$ falls strictly with A: strength does not rise and the aggregate is higher. It crosses one at most once. It is solved by bisection on $\ln A$.

The integer problem. Because $K_g^*(A)$ moves in jumps, $\mathcal{S}^*$ can jump over one. The bisection then converges to the aggregate $\bar{A}$ where some group switches rungs. If the group is assigned its larger rung, the clearing aggregate lies at or above $\bar{A}$, where the smaller rung is its best response; if it is assigned its smaller rung, the clearing aggregate lies below $\bar{A}$, where the larger rung is. Either way the group wishes to deviate, and no exact equilibrium in whole plants exists. The procedure keeps the configuration with the fewest groups wishing to deviate and reports how many there are.

8.2 General equilibrium with entry

What changes in Steps 4 and 5. Fixed costs are paid to labor where the plant is located and are deducted from the group's profit. Let Φ_g be the total fixed cost of group g's active plants, which depends on wages and plant sets only. Income becomes

$$X = wL - \theta'\Phi + \Gamma X \qquad \Longrightarrow \qquad X = (I - \Gamma)^{-1}(wL - \theta'\Phi),$$

where $\theta'\Phi$ is the vector with entries $\sum_g \theta_{gn}\Phi_g$, and each country's wage bill (23) adds the fixed costs of the plants located in it. Walras' law still holds, because every fixed cost is both a deduction from profit and a payment to labor. Scaling all wages scales fixed costs too, so it leaves every entry decision unchanged.

Two results of Steps 4 and 6 change. Income is no longer bounded below by labor income for arbitrary plant sets, because a group's profit net of fixed costs can be negative; existence of wages for given plant sets then needs fixed costs small enough that every country's income is at least a fixed fraction of its labor income. And in the hypotheses of Theorem 1, payments for fixed costs enter the demand term with a zero response, because a fixed cost paid in country n does not change when another country's wage rises.

The outer loop. Plant sets and wages are solved by an outer loop:

1. start from a configuration of plant sets, for example every potential plant active in every market of its sector;
2. given the plant sets, solve Steps 2–6 for wages, and record in every market the delivered costs, fixed costs, spending and aggregate A;
3. in every market, compute each group's ladder and best response at the recorded aggregate, and its gain from switching;
4. if no group gains, stop: every market is in an entry equilibrium at the equilibrium wages;
5. otherwise, let the groups with the largest gains switch, and return to 2.

Moving only the groups with the largest gains damps the loop. Moving every group at once can make the configuration oscillate, because wages computed for many active plants differ from wages for few. The choice affects the path of the algorithm, not its fixed points, and imposes no order of entry. The loop is not guaranteed to converge; when it does not reach a configuration in which no group wishes to move, it returns the one with the fewest such groups. Because the joint equilibrium of wages and plant sets is not known to be unique, the procedure is run from several starting points.